

Week 7 Workshop Part 2

Question 1

Using the Vanguard VBK (Small-Growth) from Week 6 Workshop, compute the variance ratio (VR) 2-period VR_2 , VR 3-period VR_3 and VR 4-period VR_4 . For each case, report the value of the variance ratio and briefly interpret your result.

Question 2

Suppose the log of the S&P500 index (p_t for $t \geq 1$) follows a random walk with drift model.

(a) Write down the model.

(b) Using the method of successive substitution, show that

$$p_t = \mu t + \varepsilon_t + \cdots + \varepsilon_1 \quad (1)$$

where ε_t is a white noise process with mean 0 and variance σ^2 .

(c) The sp500.xlsx contains p_t from Jan 1990 to Jan 2001. Report the correlogram (acf) through lag 24 of p_t . What does it tell you?

Method of successive substitution - step-by-step. To derive (1), using $p_{t-1} = \mu + p_{t-2} + \varepsilon_{t-1}$ to give

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This can be repeated using $p_{t-2} = \mu + p_{t-3} + \varepsilon_{t-2}$ to give

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This can be repeated any number of times, say n times, to give

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For simplicity, suppose that $n = t$

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as required, assuming that $p_0 = 0$.